DOMESTIC AIRLINES — TRAINING

PBN Refresher

Course handbook

Performance-based navigation: RNAV and RNP, GNSS integrity and interference, the navigation database, RNAV procedures and RNP approaches.

5 lessons · 2 modules · about 0.5 hours

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.

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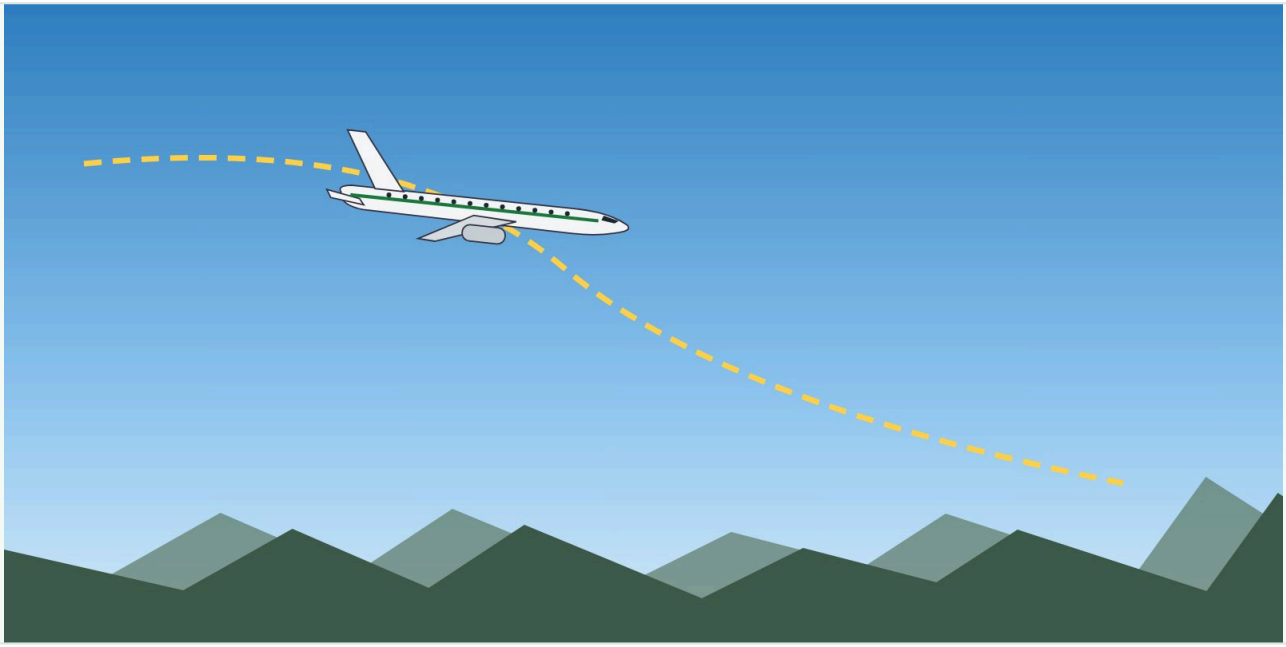
MODULE 1 · PBN PRINCIPLES

Lesson 01: The PBN Concept

Lesson 1 of 5 · about 7 min

LEARNING OBJECTIVES

- Explain the performance-based navigation (PBN) concept.
- Distinguish RNAV and RNP specifications.
- Identify the elements of a PBN operation.



PBN: the performance matters, not the sensor.

From sensors to performance

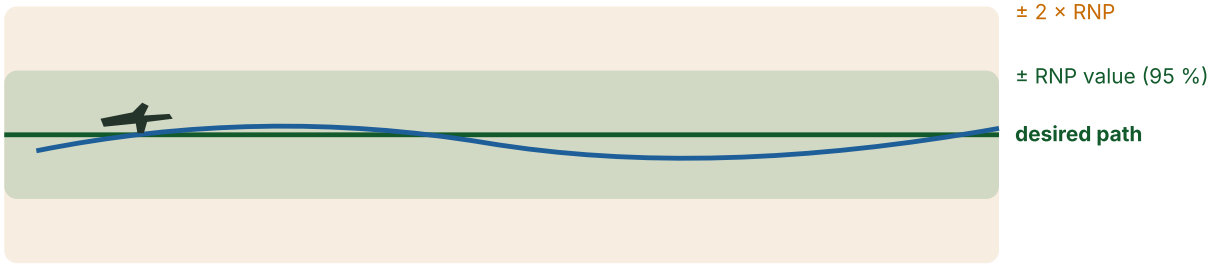
Conventional navigation is based on flying to and from ground beacons (VOR, NDB). **Performance-based navigation (PBN)** (ICAO Doc 9613) defines instead the **performance** an aircraft's area navigation system must achieve — accuracy, integrity, continuity and functionality — for a given airspace or procedure.

A PBN operation has three elements:

- the **navigation specification** — the aircraft, system and crew requirements;
- the **navaid infrastructure** — GNSS, DME/DME, VOR/DME;
- the **navigation application** — the routes and procedures designed for it.

RNAV and RNP

RNP: the aircraft must stay within ± the RNP value at least 95 % of the flight time —
and an RNP system must monitor its own performance and alert the crew when it cannot.



actual flight path = path definition error + flight technical error + navigation system error

Total system error combines the path definition, flight technical and navigation system errors. On-board performance monitoring and alerting is what makes a specification “RNP”.

- An **RNAV** specification has no requirement for on-board performance monitoring and alerting.
- An **RNP** specification **requires on-board performance monitoring and alerting** — the system tells the crew when it cannot meet the required performance (for example “UNABLE REQD NAV PERF – RNP” or a similar alert).
- The number (e.g. RNP 1) is the lateral accuracy in nautical miles that must be achieved at least 95 % of the flight time.

Navigation specifications

Phase	Typical specifications
Oceanic / remote	RNAV 10, RNP 4, RNP 2
En route continental	RNAV 5 (B-RNAV), RNAV 2, RNP 2, Advanced RNP
Terminal (SID / STAR)	RNAV 1, RNP 1, Advanced RNP
Approach	RNP APCH (LNAV, LNAV/VNAV, LP, LPV), RNP AR APCH

The aircraft's **AFM / AFM supplement** states which specifications it is certified for, and the **MEL** gives the dispatch requirements. Check your fleet's capabilities in the OM B.

KEY TAKEAWAYS

- PBN = navigation specification + navaid infrastructure + navigation application.
- RNP requires on-board performance monitoring and alerting; RNAV does not.
- RNP 1 = ±1 NM at least 95 % of the time.
- Fleet capability: AFM, OM B and MEL.

KNOWLEDGE CHECK

What distinguishes an RNP specification from an RNAV specification?

A. A smaller number

B. On-board performance monitoring and alerting

C. The use of VOR

D. Autopilot coupling

Answer: B. RNP requires on-board performance monitoring and alerting.

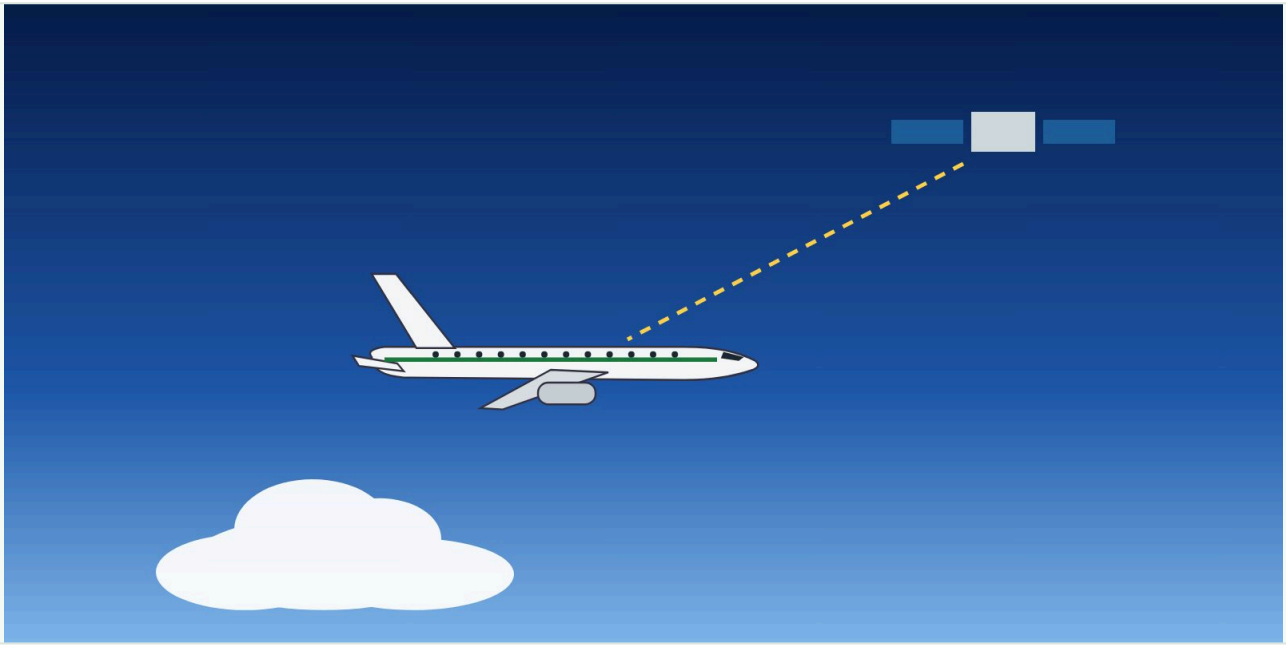
MODULE 1 · PBN PRINCIPLES

Lesson 02: GNSS, Integrity and the Navigation Database

Lesson 2 of 5 · about 7 min

LEARNING OBJECTIVES

- Explain GNSS integrity monitoring and its limits.
- Recognise GNSS interference threats.
- Check and use the navigation database correctly.



GNSS integrity — and interference — must be monitored.

GNSS and integrity

- Most PBN operations rely on **GNSS**. Integrity is provided by **receiver autonomous integrity monitoring (RAIM)** or equivalent aircraft-based augmentation, and by satellite-based augmentation (SBAS) where available.
- Check the **GNSS NOTAMs** and the predicted RAIM availability for the route and the time of the approach, as the company procedure requires.
- Know the backup: DME/DME or IRS updating, and conventional procedures and navaids.

GNSS interference

Jamming (signal blocked) and **spoofing** (false signal) are increasingly common in some regions. Effects include loss of GNSS position, position or time jumps, EGPWS terrain alerts, ADS-B and clock errors. Follow the company and manufacturer guidance: cross-check position with other sources, consider deselecting GNSS where the procedure allows, use conventional navaids, inform ATC and report.

The navigation database

- Check the **AIRAC cycle** is current before flight.
- Procedures must be **loaded from the database** by name — never built manually from waypoints for an RNP approach.
- Cross-check the loaded procedure with the chart: waypoint sequence, tracks, distances, altitude and speed constraints, and the vertical path angle on the final approach.
- Do not modify the final approach segment of an RNP approach (except where your procedure allows specific changes).

KEY TAKEAWAYS

- Check GNSS NOTAMs and RAIM / integrity availability.
- Jamming and spoofing: cross-check, use conventional navaids, inform ATC and report.
- Current AIRAC; load procedures by name; cross-check with the chart; do not modify the final approach.

KNOWLEDGE CHECK

How must an RNP approach be loaded in the FMS?

A. Built from individual waypoints

B. Loaded by name from the current navigation database and cross-checked with the chart

C. Copied from the previous flight

D. Entered as a direct-to

Answer: B. RNP approaches must be loaded from the database and checked against the chart.

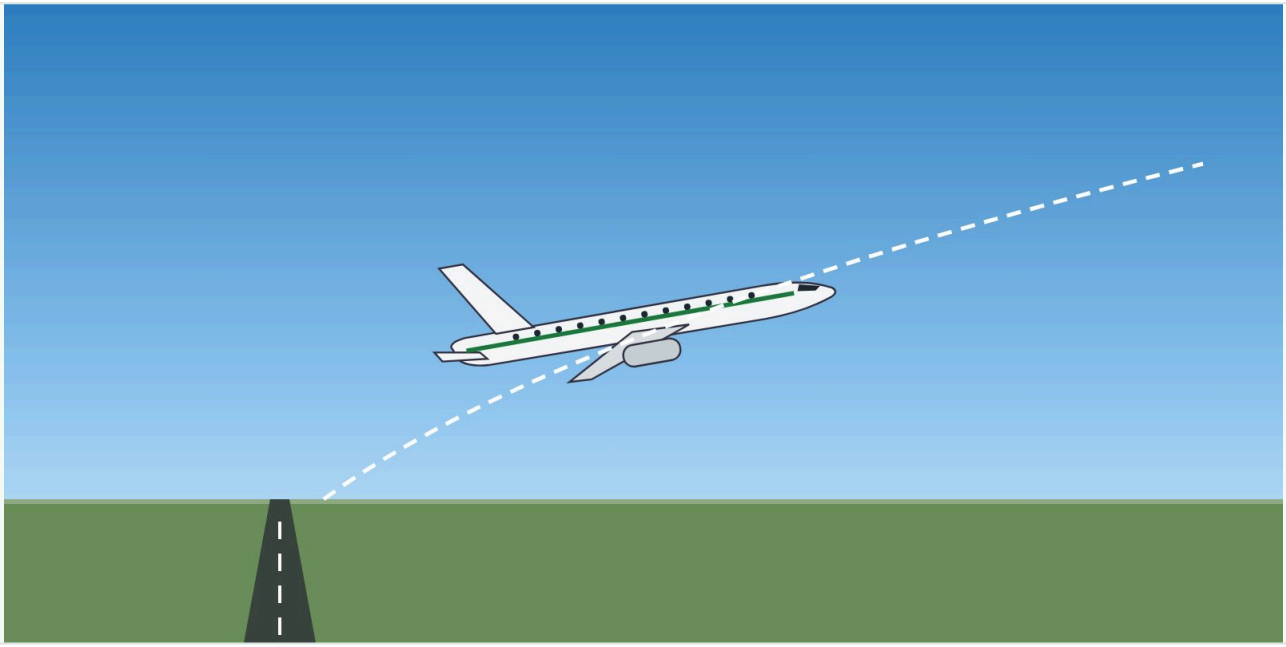
MODULE 2 · PBN OPERATIONS

Lesson 03: RNAV Departures, Arrivals and En-route Operations

Lesson 3 of 5 · about 6 min

LEARNING OBJECTIVES

- Fly RNAV SIDs and STARs correctly.
- Apply flight technical error limits.
- Manage contingencies on RNAV routes.



Fly the SID with LNAV and respect the constraints.

Before departure

- Check that the aircraft and crew are approved and the equipment required by the MEL is operational.
- Load and cross-check the SID: waypoints, fly-by and fly-over points, altitude and speed constraints.
- Brief the procedure and any changes that ATC may give (e.g. direct-to clearances).

Flying the procedure

- Use the lateral navigation mode (LNAV) and the flight director or autopilot, as the SOP requires, to keep the **flight technical error** small — typically within half the navigation accuracy (for RNAV 1 / RNP 1: within 0.5 NM).
- Monitor the cross-track deviation and the navigation performance indications.
- Respect the published altitude and speed constraints — they are part of the procedure.

Contingencies

- If the navigation performance cannot be met (alert, loss of GNSS, system failure), **inform ATC** — “unable RNAV due equipment” or the appropriate phrase — and request an alternative clearance.
- Follow the QRH / FCOM procedure for the navigation failure.
- Report the event.

KEY TAKEAWAYS

- Load and cross-check the SID or STAR, including constraints.
- Keep flight technical error within half the navigation accuracy.
- If performance is lost, tell ATC and request another clearance.

KNOWLEDGE CHECK

On an RNAV 1 SID, the lateral flight technical error should normally be kept within:

A. 0.1 NM

B. 0.5 NM

C. 2 NM

D. 5 NM

Answer: B. Typically within half the navigation accuracy: 0.5 NM for RNAV 1.

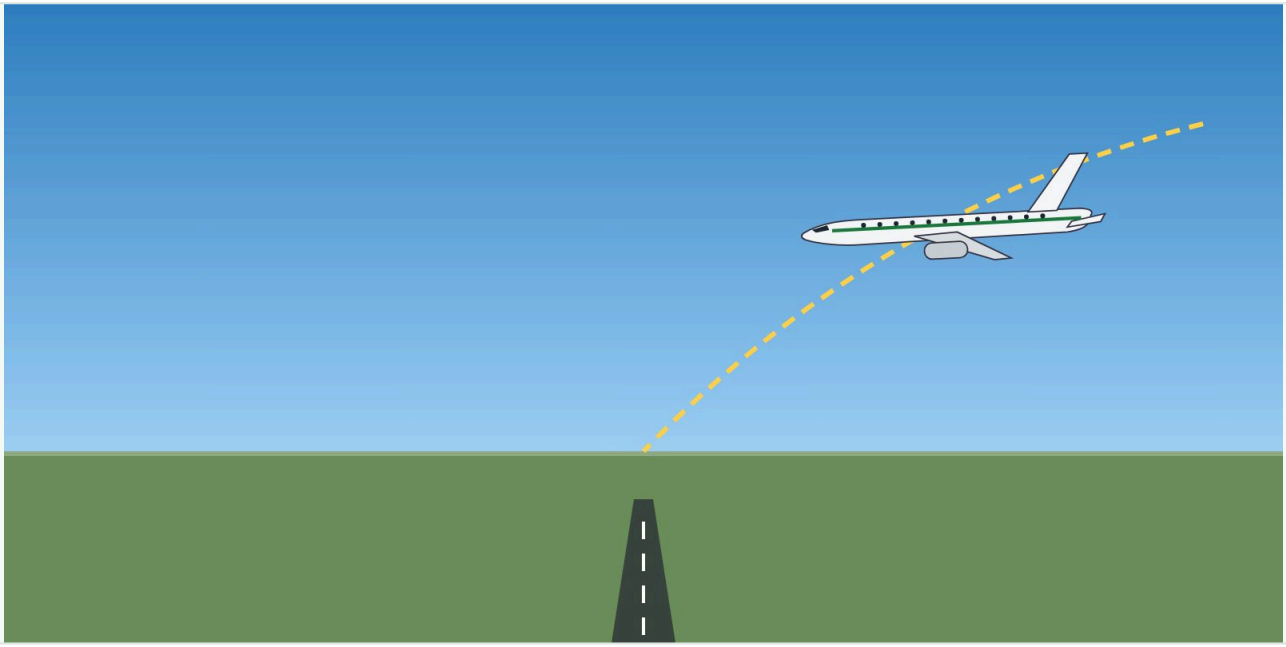
MODULE 2 · PBN OPERATIONS

Lesson 04: RNP Approaches: LNAV, LNAV/VNAV, LPV

Lesson 4 of 5 · about 8 min

LEARNING OBJECTIVES

- Distinguish the RNP APCH minima lines.
- Apply the temperature limits for baro-VNAV.
- Fly RNP approaches with the correct technique and go-around criteria.



LNAV, LNAV/VNAV or LPV: fly the minima you are authorised for.

Minima lines on an RNP chart

Minima	Guidance	Type
LNAV	Lateral only — flown with CDFA to an MDA	2D
LNAV/VNAV	Lateral + barometric vertical guidance (baro-VNAV) to a DA	3D
LP	Lateral SBAS guidance to an MDA	2D
LPV	Lateral + vertical SBAS guidance to a DA	3D

Use only the minima line your aircraft is certified and the company is authorised for. In EASA rules, **RNP APCH** (LNAV, LNAV/VNAV, LP, LPV) does not need a specific approval, but crews must be trained; **RNP AR APCH** requires a specific approval (SPA.PBN).

Temperature and baro-VNAV

- In cold air, the baro-VNAV path is **lower and shallower** than published; in hot air it is higher and steeper.

- Charts with LNAV/VNAV minima state a **temperature limit** — below it, LNAV/VNAV must not be used (use LNAV minima or another approach).
- Check the aerodrome temperature against the chart before every LNAV/VNAV approach.

Technique

- Brief the approach with the chart and the FMS: minima, vertical path angle, temperature limit, missed approach.
- Be stabilised by the company gates; use the autopilot and the appropriate vertical mode as your SOP requires.
- Monitor lateral and vertical deviations. Go around if the deviation limits in your SOP are exceeded, or on any navigation alert (e.g. loss of integrity) — unless the required visual references are established.
- The missed approach is part of the RNP procedure — fly it in LNAV as published.

KEY TAKEAWAYS

- LNAV and LP = 2D to an MDA; LNAV/VNAV and LPV = 3D to a DA.
- RNP APCH: no specific EASA approval but training required; RNP AR: specific approval.
- Respect the chart temperature limit for baro-VNAV.
- Go around on excessive deviation or navigation alert.

KNOWLEDGE CHECK

The OAT is below the minimum temperature published for LNAV/VNAV on the chart. What do you do?

A. Fly LNAV/VNAV anyway

B. Do not use LNAV/VNAV minima — use LNAV or another approach

C. Increase the DA by 50 ft

D. Fly faster

Answer: B. Below the temperature limit the baro-VNAV path is not protected.

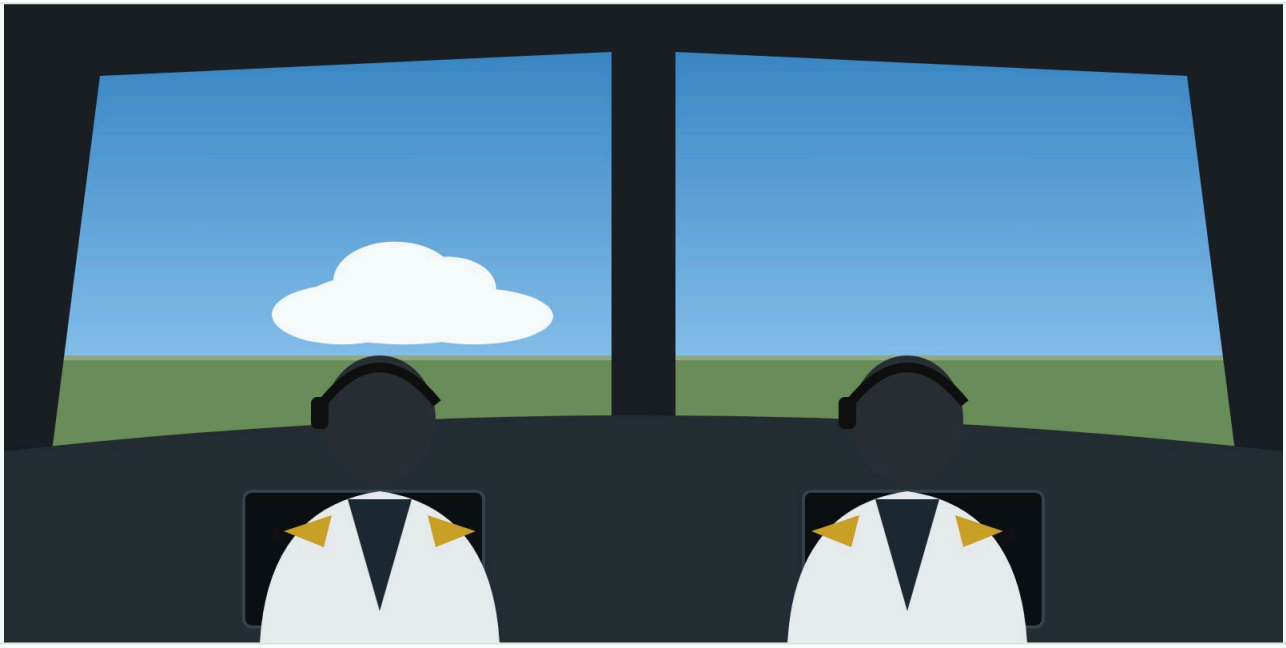
MODULE 2 · PBN OPERATIONS

Lesson 05: Approvals, MEL and Crew Responsibilities

Lesson 5 of 5 · about 5 min

LEARNING OBJECTIVES

- Identify the approvals and documents for PBN operations.
- Apply the MEL to navigation equipment.
- Summarise crew responsibilities in PBN operations.



Plan, check, fly, monitor, act, report.

Documents and approvals

- The AFM states the aircraft's PBN capabilities. The OM B lists the company's PBN operations by fleet.
- The flight plan item 10 and item 18 (`PBN/`) codes must match the aircraft's actual capability and approvals.
- Specific approvals (for example RNP AR APCH) are listed in the operations specifications of the AOC.

MEL

The MEL includes the dispatch requirements for systems that need a specific capability or approval — RNAV/PBN, RVSM and LVO. Check the MEL for FMS, GNSS receivers, IRS and display units before planning a PBN route or approach.

Crew responsibilities

1. Plan: NOTAMs, GNSS integrity, database currency, equipment and MEL.

2. Load and cross-check procedures from the database.
3. Fly with the required modes and monitor deviations and alerts.
4. Act on alerts: inform ATC, apply the contingency, go around when required.
5. Report navigation errors and GNSS interference.

KEY TAKEAWAYS

- AFM + OM B + AOC specify PBN capability and approvals; flight plan codes must match.
- Check the MEL for FMS, GNSS, IRS and displays.
- Plan, load and check, fly and monitor, act on alerts, report.

KNOWLEDGE CHECK

Where are specific approvals such as RNP AR APCH recorded?

A. In the AFM only
B. In the operations specifications of the AOC
C. In the flight plan
D. On the chart

Answer: B. Specific approvals appear in the operations specifications.

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